Tutorial No : AMVS 4

Topic : PIPE SIZING

1. State the two main functions of a chilled water expansion tank. [4 points]

2. Figure 2 shows a simple chilled water circuit. Describe how the chilled water flow will change throughout the system when the cooling demand at the cooling coils decreases. You may start by assuming that the cooling coils are operating at full load and that there is no water flow through the bypass line. You may assume that the two chillers are identical and so are the air handling units. [8 points]

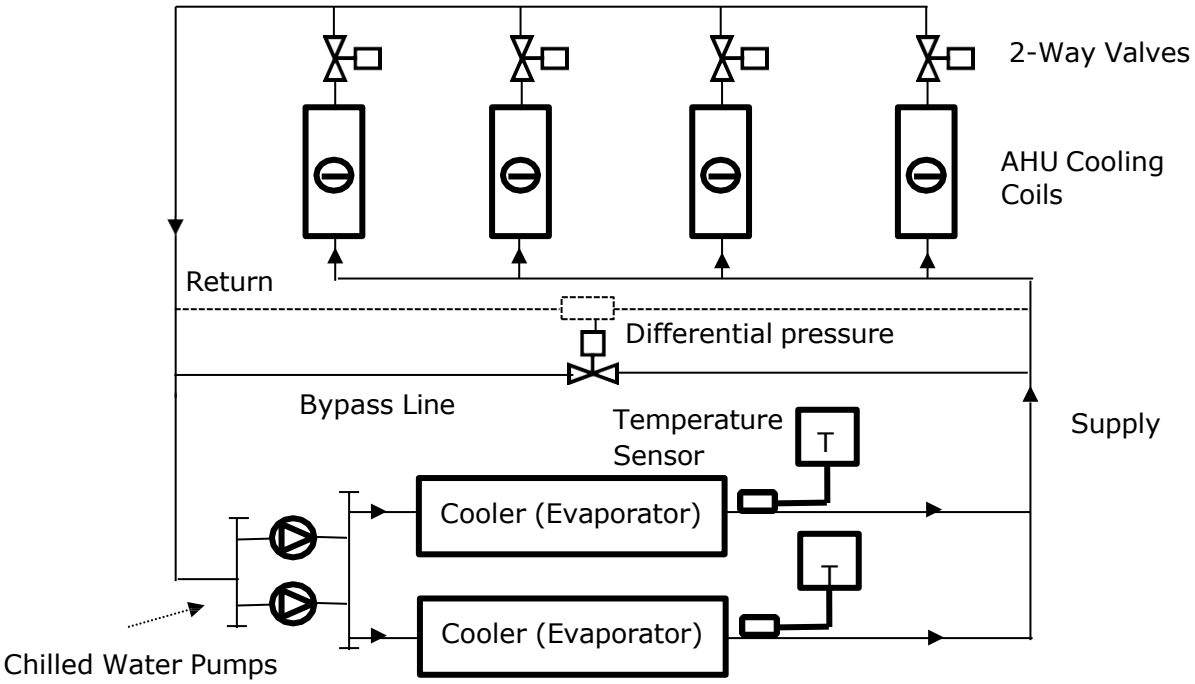


Figure 2

3. Figure 3 shows the schematic diagram of the chilled water circuit of an air-conditioning system. The chilled water pumps are identical and only one would be in operation at any one time. Size the index run pipe sections and the required chilled water pump head using the equivalent length method and the information provided. Assume a safety factor of 10%. [25 marks]

Legend

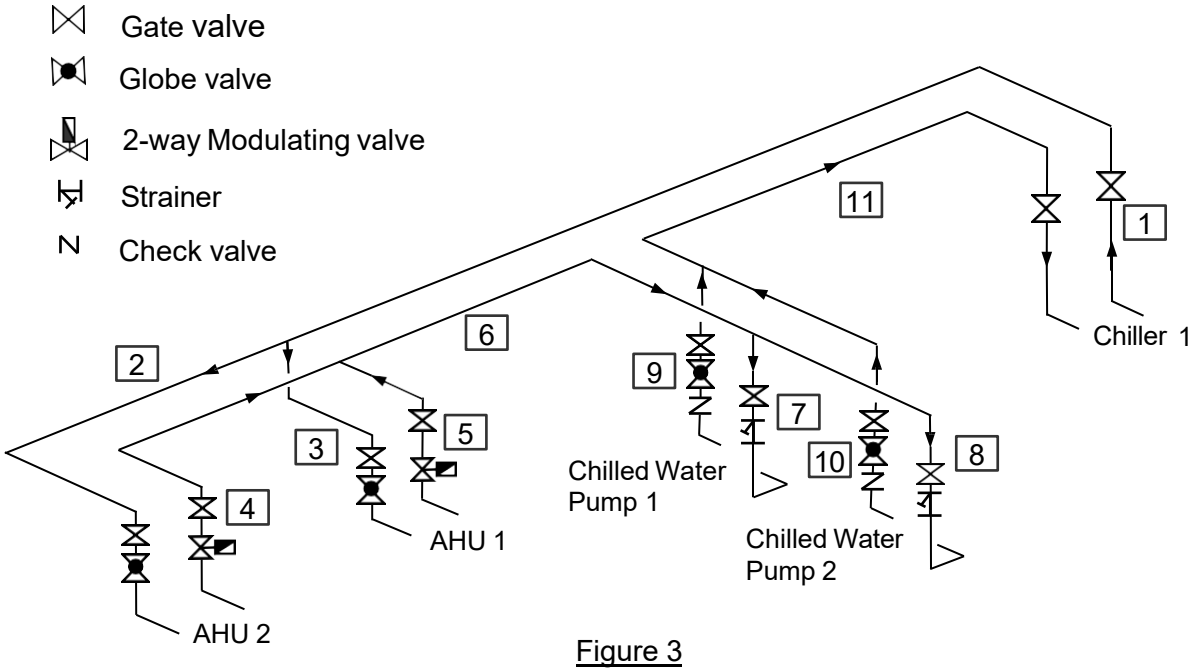


Figure 3

Pipe Length

Section	1	2	3	4	5	6	7	8	9	10	11
Pipe Length (m)	20	16	6	15	5	15	3	5	3	5	10

Pressure Drop

Equipment	Chilled Water Volume Flow Rate (l/s)	Pressure Drop (kPa)
Chiller	To determine	60
AHU 1	5	25
AHU 2	5	30

Valves and Pipe Fittings L_e/D

Valves and Pipe Fittings	L_e/D
Gate valve	7.5
Globe valve	320
Check valve	95
Strainer	250

Valves and Pipe Fittings	L_e/D
2-way modulating valve	37
Elbow	55
Tee - straight thru'	20
Tee - branch	60

Assume losses due to reducers and enlargers are negligible.

Range of acceptable pressure loss/m : 100 to 400 Pa/m

4. Figure 4 shows the schematic diagram of the condenser water circuit of an air-conditioning system with identical cooling towers and condenser water pumps. During normal operation, the chiller, **both** cooling towers and **one** condenser water pump are in operation. The pressure loss of Section 1 of the index run is already calculated and shown in the pipe sizing tables provided. Size the remaining sections of the index run using the pipe friction loss chart and use the equivalent length method to determine the required condenser water pump head. Assume a safety factor of 10%. [25 marks]

Section	Pipe Length (m)	Section	Pipe Length (m)
1	20	**6**	8
2	4	**7**	4
3	8	**8**	8
4	5	**9**	4
5	15	**10**	15

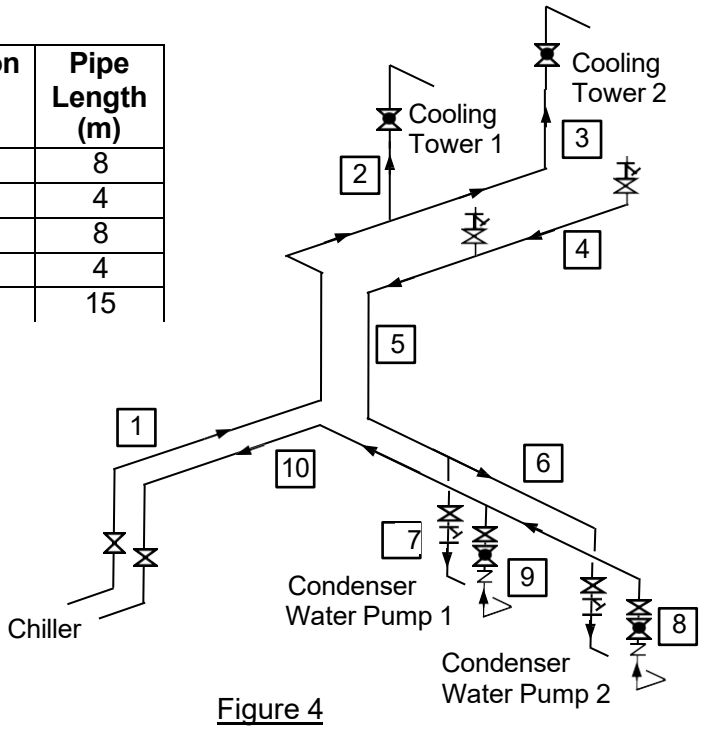


Figure 4

Pressure Drop

Equipment	Condenser Water Volume Flow Rate (l/s)	Pressure Drop (kPa)	Static Head (m of water)
Chiller	To determine	30	-
Cooling Tower 1	10	5	3
Cooling Tower 2	10	5	3

Valves and Pipe Fittings L_e/D

Valves and Pipe Fittings	L_e/D
Gate valve	7.5
Globe valve	320
Check valve	95
Strainer	250

Valves and Pipe Fittings	L_e/D
2-way modulating valve	37
Elbow	55
Tee - straight thru'	20
Tee - branch	60

Assume losses due to reducers and enlargers are negligible.
Range of acceptable pressure loss/m: 100 to 400 Pa/m

Pipe Sizing Table (Equivalent Length) for Question 4

System	Chilled Water Circuit/Condenser Water Circuit* *Delete accordingly				Index Run				
Section	Volume Flow Rate (l/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
1									**9,828**
								Sub-total to be carried forward	

Pipe Sizing Table (Equivalent Length) (Contd)

Section	Volume Flow Rate (l/s)	Pipe Dia. (mm)	Water Velocity (m/s)	Fittings and Valves	No. of Fittings or Valves	Equivalent Length (m)	Total Equivalent Length (m)	Pressure Loss (Pa/m)	Section Pressure Loss (Pa)
						Sub-total carried forward from the previous page			
Working Space for Pump Head Calculations:								Total Pressure Loss	
								Safety Factor (%)	
								Grand Total Pressure Loss	
								Frictional Head (m)	
								Static Head (m)	
								Pump Head (m)	